

Inspect AI: Application User Manual & Feature Flow

What is Inspect AI?

Inspect AI is an automated, AI-driven computer vision application designed for quality control in pharmaceutical manufacturing. It acts as an intelligent digital inspector that analyzes images of pill bottles in real-time to detect microscopic manufacturing defects (like surface contamination, dirt, or structural fractures).

Powered by a state-of-the-art anomaly detection AI (PatchCore) and integrated with simulated enterprise systems (like SAP Quality Management), the app completely removes the need for manual, eye-straining inspections by instantly routing bottles into three categories: **Conforming (Pass)**, **Human Review (Hold)**, or **Defected (Reject)**.

Application Walkthrough & Features

1. The Initial State (System Ready)

When you first open the application, you are greeted by the main dashboard.

- **The Center Console:** Displays a sleek, professional "System Ready" empty state. It confirms that the AI model is active and awaiting sensor input.
- **The Left Panel ("Sample Bottles"):** This is the core navigation panel. It contains a randomly shuffled queue of over 80 dataset images pulled directly from the inspection line.
- **The Notification Bell (Top Right):** A real-time alerts hub that flashes a red badge counter whenever a defect is detected or a batch finishes processing.

2. Single Image Inspection (The Diagnostic Report)

When you click on any single image from the left panel's list, the AI processes that specific bottle and opens the detailed **Diagnostic Report** page. This page is divided into multiple sections:

A. The Verdict Banner (Top Section)

A prominent banner that gives you an immediate summary of the AI's decision.

- **Verdict Status:** Displays the final outcome (e.g., "VERDICT: AUTO-REJECT" with a message like "REJECTED — Container visual defects exceed allowed limit").
- **AI SCORE:** The precise mathematical deviation value calculated by the neural network for this specific bottle (e.g., 16.80).
- **STANDARD LIMIT:** The acceptable baseline threshold (default is **10.46**).
 - If AI Score is < 10.46 , it passes automatically (Auto-Pass).
 - If it's between 10.46 and 13.50, it requires human review (Grey Zone).
 - If it's ≥ 13.50 , it is automatically rejected (Auto-Reject).
- **SCAN TIME:** How fast the AI processed the image in milliseconds (e.g., 1036 ms).

B. The Visualizer Section (Middle Section)

This is where Explainable AI (XAI) is showcased, allowing the user to see exactly *what* the AI is detecting.

- **Tabs:** You can switch between four different visual views:
 - **BBox (Bounding Box):** Shows the raw camera feed with a precise red box drawn around any detected defects.
 - **Heatmap:** A thermal overlay. Cold colors (blue) mean the bottle is normal, while hot colors (red/yellow) indicate structural anomalies.
 - **Blended:** A transparent mix of the heatmap over the original image to easily pinpoint where the anomaly exists on the bottle.
 - **Multi-Panel:** A comprehensive presentation card showing the original, masks, and heatmaps side-by-side.
- **Sensitivity Slider (Sens):** A slider located on the right side of the tabs. Users can manually drag this to adjust the "Standard Limit" threshold dynamically. Adjusting this will instantly recalculate if the bottle passes or fails based on the new custom limit.

C. Human Approval & Override (For "Grey Zone" Items)

If a bottle falls into the borderline **Grey Zone**, the application suspends auto-routing and prompts for human intervention.

- **Override Panel:** A dedicated module appears allowing the QA Inspector to manually override the AI.

- **21 CFR Part 11 Compliance:** To "Approve" or "Reject" the bottle, the inspector must enter their Username, Password, and a Justification Reason.
- **Audit Logging:** Submitting this form applies the override and logs a permanent, timestamped digital signature into the system's simulated SAP audit trail.

D. Inspection Details (Bottom Section)

A structured metadata table that maps the AI's visual findings directly into simulated enterprise actions.

- **Defect Class ID:** Categorizes the type of defect found (e.g., DC-V04 (Fractured Casing) or DC-V01 (Surface Dirt)).
- **Severity:** Ranks the critical nature of the defect (e.g., MAJOR / CRITICAL or MINOR).
- **Pass / Fail:** The final status assigned to the bottle.
- **SAP Action:** Displays the specific ERP / Quality Management protocol that is triggered. For example, a failing bottle will trigger QE11 -> FAIL, BAPI_NOTIF_CREATE (which simulates posting a fail result and creating a quality deviation notification in an SAP system).

3. Notifications & Real-Time Alerts

To ensure QA managers never miss a critical failure, a live notification system operates in the background.

- **The Bell Dropdown:** Clicking the bell icon in the top header opens an interactive dropdown menu.
- **Live Event Logging:** Any time a bottle is flagged as a "Reject" or enters the "Grey Zone", a distinct alert is pushed to this dropdown with the exact timestamp and score.
- **Clear Alerts:** Users can click the "Clear" button to wipe the notification history once issues are acknowledged.

4. Batch Processing ("Inspect All Images")

Instead of clicking images one by one, the application can process the entire manufacturing run at once.

- Located right at the top of the left panel is a large blue **INSPECT ALL IMAGES** button.
- When clicked, the application shifts into high-speed batch mode:
 - **The Progress Dashboard:** The center screen is replaced by a live Batch Inspection Dashboard. A progress bar rapidly fills up as the AI concurrently

processes all 80+ images in the background using a highly optimized "Fast-Mode".

- **Live Analytics:** Three large counter boxes appear at the top, tallying up the exact number of **Good/Conforming**, **Human Review Needed**, and **Defected** bottles in real-time.
- **Batch Alerts:** Upon completion, the system automatically fires a summary alert into the Notification Bell (e.g., *"Batch Inspection Complete: 5 Rejects, 2 Reviews detected"*).
- **The Results Grid:** Below the analytics, a grid populates with every single bottle processed. Each bottle is displayed as a clean card tinted Green, Grey, or Red depending on its final status.

5. Drill-Down Investigation

The batch results grid isn't just a static report—it's fully interactive.

- If a QA manager notices a Red (Fail) card in the batch grid and wants to know *why* it failed, they can simply click directly on that card.
- **Seamless Transition:** Clicking a card instantly pulls up that specific bottle's full, high-resolution Diagnostic Report (complete with the heatmaps, override tools, and SAP logs mentioned in Step 2).
- Once the investigation is complete, the user can click the "Back" arrow in the top header to return directly to the empty state and start a new inspection.